Present Day, Present Time: AI Exposure & Labor Market Risk

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The Question

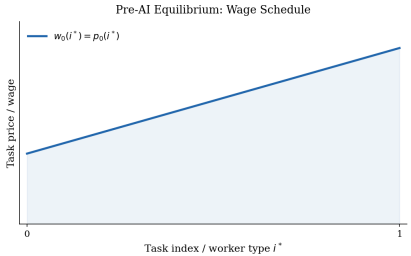
November 2022: ChatGPT launches.

Within months, millions of people are using AI to:

- Draft emails and documents
- Write and debug code
- Summarize and analyze text
- Answer legal and financial questions

These are jobs people get paid to do.

Did it cost them?



Some workers are more exposed than others

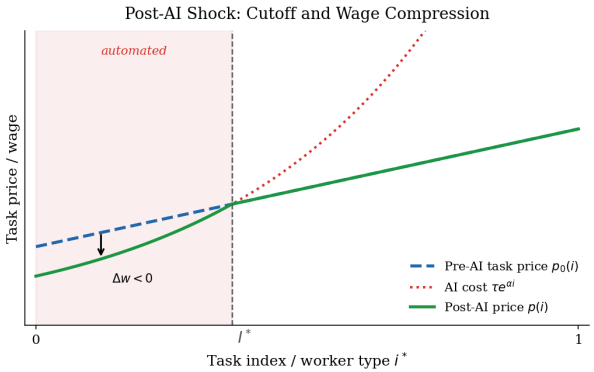
A Lesson from Trade Economics

Offshoring (2000s): companies moved jobs abroad when foreign labor was cheaper

- **Grossman & Rossi-Hansberg (2008):** not whole jobs — specific *tasks* get offshored
  - Data entry → India. Customer calls → Philippines. R&D → stays home.
  - Workers in offshored tasks faced wage cuts or job loss
- **Ottaviano, Peri & Wright (2013):** different workers face different cutoffs — not everyone equally at risk
- **AI is the same story, new technology:**
  - Instead of cheaper foreign workers — a cheaper *machine*
  - **"Offshoring" your tasks to Claudeland, GPTopia, Geminia...**
  - Workers whose tasks AI can undercut face the same pressure

The Model: Who Gets Hit?

Key idea: rank every cognitive task by how easy it is for AI to do



- Tasks on the left: **routine** — AI can do them cheaply
  - drafting, summarizing, basic coding
- Tasks on the right: **complex** — AI still struggles
  - strategy, judgment, novel problem-solving
- **Cutoff I^*:** the point where AI cost = human wage
  - Left of I^*: AI undercuts → wage falls
  - Right of I^*: humans retain edge

What the Model Predicts

Workers to the left of l^* face two immediate pressures:

Wage compression

Firms can switch to AI $\rightarrow$ workers must accept lower pay to stay employed

$$\Delta w < 0 \quad \text{for exposed workers}$$

The more exposed your tasks, the steeper the cut

Displacement

If the compressed wage drops below what you're willing to accept $\rightarrow$ you exit

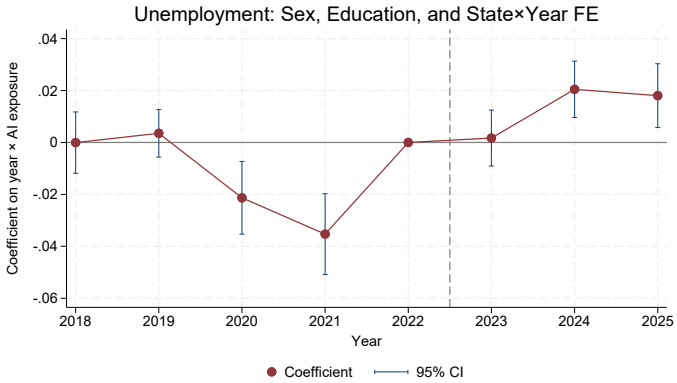
$$w_{\text{compressed}} < \underline{w} \;\Rightarrow\; \text{unemployed}$$

Displacement is the extreme case of wage compression

Note: this captures the **immediate, substitution effect** *only. Longer-run channels (AI augmenting workers, creating new tasks, expanding output) are real but operate over a longer horizon — and could go the other way.*

Do We Actually See This?

Data: CPS survey, 500K+ U.S. workers, 2018–2025
AI exposure score: Eloundou et al. (2024) — occupation-level LLM exposure
Strategy: compare high- vs. low-exposure workers before and after ChatGPT (2022)



- Flat pre-trends before 2022 — comparable groups
- **Post-2022:** high-exposure workers show a clear, rising unemployment gap
- Consistent with the model: workers left of I^* are getting displaced
- Effect is stable across all specifications

Thank you!

Questions welcome.

Bottom line:

AI is like offshoring —
some workers face immediate
wage and employment risk

The data says:

Yes, high-exposure workers
are already seeing it
in unemployment rates

Backup: The Dirty Work

Formal model, derivations, and further discussion of I^*

Model Setup (Formal)

- **Tasks** indexed $i \in [0, 1]$ by AI susceptibility
  - Low i: routine cognitive — easy for AI — High i: complex — hard for AI *currently*
- **CES Production**:

$$Y = \left[\int_0^1 y(i)^{\frac{\sigma-1}{\sigma}} \, di \right]^{\frac{\sigma}{\sigma-1}}$$

- **Workers:** type i^* specializes in task i^*; supply of task i is $f(i)$ (fixed, inelastic)
- **AI cost** of task i: $c^{\mathrm{AI}}(i) = \tau \cdot e^{\alpha i}$
  - Lower τ = more capable AI; $e^{\alpha i}$ ensures complex tasks remain costly for AI
- Competitive firms; output price P_Y fixed (small open economy)

Pre-AI Equilibrium: Derivation

Firm cost minimization gives task demand:

$$y^d(i) \;=\; Y \cdot \left(\frac{P_Y}{p_0(i)}\right)^{\sigma}$$

Market clearing $\left(y^d(i) = f(i)\right)$ pins down task price:

$$p_0(i) \;=\; P_Y \cdot \left(\frac{Y}{f(i)}\right)^{1/\sigma}$$

Solve for Y_0 — plug $y(i) = f(i)$ back into the CES aggregator:

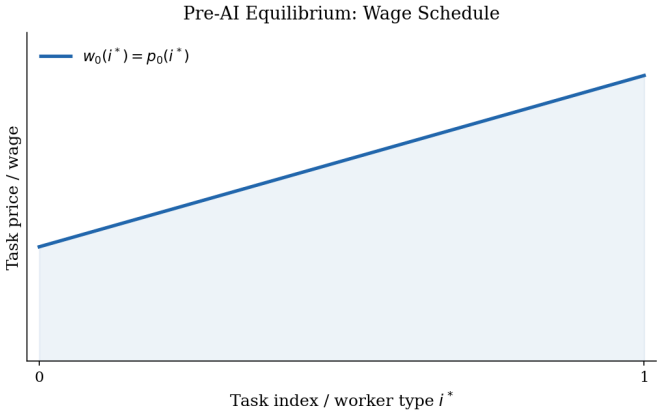
$$\boxed{Y_0 \;=\; \left[\int_0^1 f(i)^{\frac{\sigma-1}{\sigma}}\, di\right]^{\frac{\sigma}{\sigma-1}}}$$

- Y_0 is fully determined by worker distribution f — not by prices
- Y_0 is the output level held fixed in the short-run AI shock analysis

Pre-AI Equilibrium: Wage Schedule

Under full specialization, each worker earns their task price:

$$w_0(i^*) \;=\; p_0(i^*) \;=\; P_Y \cdot \left(\frac{Y_0}{f(i^*)}\right)^{1/\sigma}$$



- Each worker earns the competitive price of their task
- Scarce types earn more; abundant types earn less

Upward slope for illustration only — no shape restriction on p_0

Immediate Effects: What We Do and Don't Model

Hold $Y = Y_0$ fixed — immediate impact before the economy adjusts

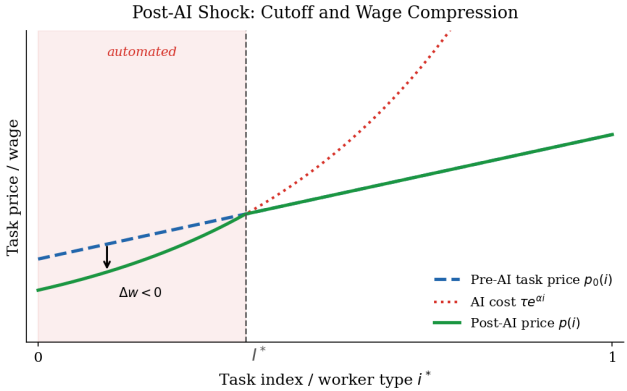
Isolates the **substitution channel** only. Longer-run channels set aside:

- **Augmentation** — AI complements some workers, raising productivity over time
- **Output expansion** — cheaper production eventually raises Y beyond Y_0
- **Task creation** — AI creates new tasks over time (Acemoglu & Restrepo 2019)

Immediate task price once AI enters:

$$p(i) = \min\big(p_0(i),\ \tau e^{\alpha i}\big) \qquad \Rightarrow \qquad \text{cutoff } I^* :\ \tau e^{\alpha I^*} = p_0(I^*)$$

Immediate Effects: Cutoff and Wages



- $i < I^*$: AI immediately undercuts → wage compression

$$\Delta w = \tau e^{\alpha i^*} - p_0(i^*) < 0$$

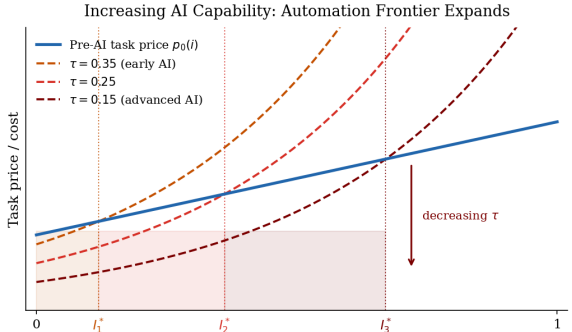
- $i \geq I^*$: no direct price pressure in the short run
  - ↪ output expansion raises their wages over time
- Compressed wage < reservation → **exit employment**

Further Discussion of I^*

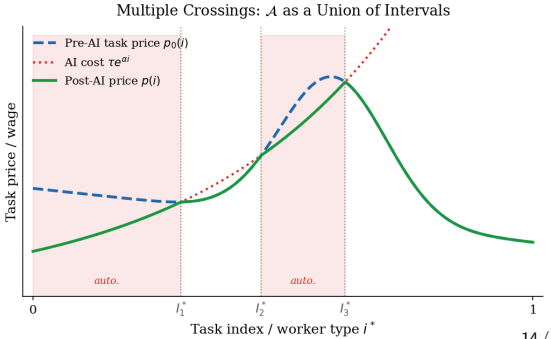
As AI improves ($\tau \downarrow$), I^* rises

$$\frac{\partial I^*}{\partial \tau} < 0$$

The automation frontier expands toward more complex tasks.



Non-monotone $p_0(i)$: multiple cutoffs

$$\mathcal{A} = [0, I_1^*] \cup [I_2^*, I_3^*] \cup \cdots$$

Wage compression applies to all $i^* \in \mathcal{A}$; exposed workers need not be at the low end.